Neural Mechanisms of Spatial Choice: dCA1 Plasticity, Cue Processing & Behavioral Variability

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INTRODUCTION

Hippocampal synaptic plasticity, particularly LTP in dorsal CA1 (dCA1), has been associated with spatial memory, but recent studies suggest a role in spatial decision-making. Spatial choices require integration of environmental cues to guide navigation. Here, we sought to better understand the dynamics of spatial choice and the role of hippocampal synaptic plasticity. In our task, mice exhibited substantial variability in performance, which we further investigated using whole-brain c-Fos mapping.

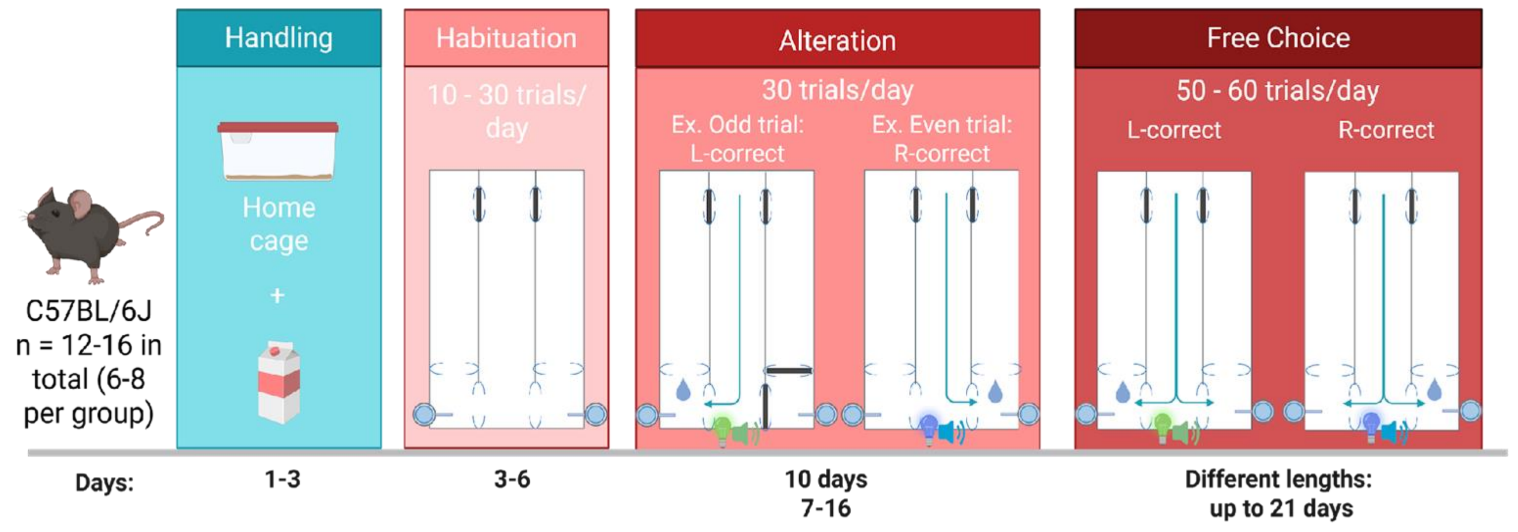
AIMS

To 1) examine the role of plasticity in spatial choices, how it affects cue processing and to 2) understand what are the neural underpinnings of performance differences in spatial choice task.

METHODS

Young adult C57BL/6J mice were trained in an automated E-maze in three independent experiments. Two experiments examined the effects of dCA1 plasticity disruption by lentiviral PSD-95 shRNA or control luciferase shRNA: one during standard training and one during cue-useage tests, in which animals were tested under three cue configurations. A separate, non-injected cohort underwent standard training followed by whole-brain c-Fos mapping (Olympus VS110, ABBA, QuPath, custom Python scripts) to identify neural correlates of individual differences in task performance.

Standard e-maze training

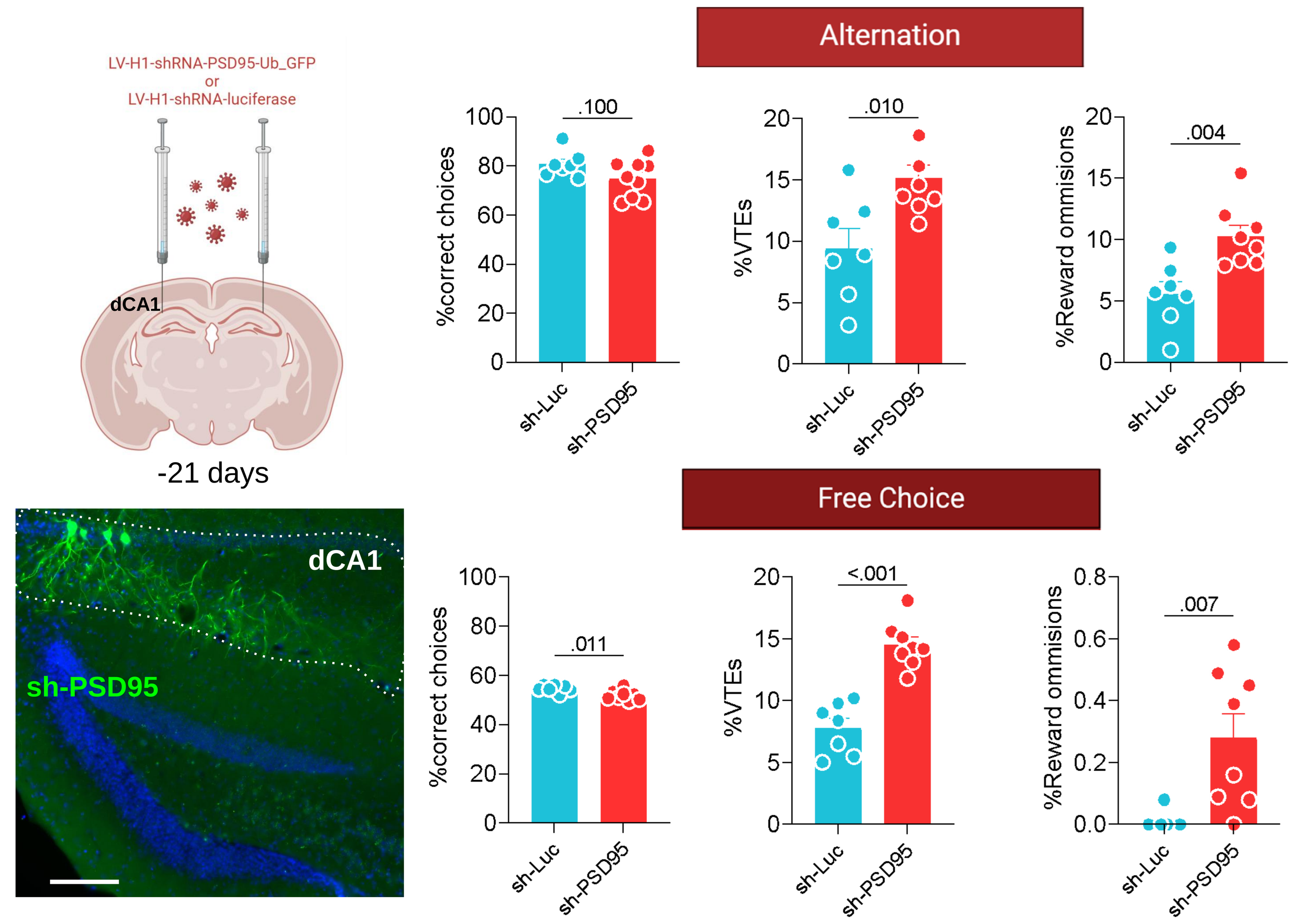


Cue usage tests

Scan me for more details!

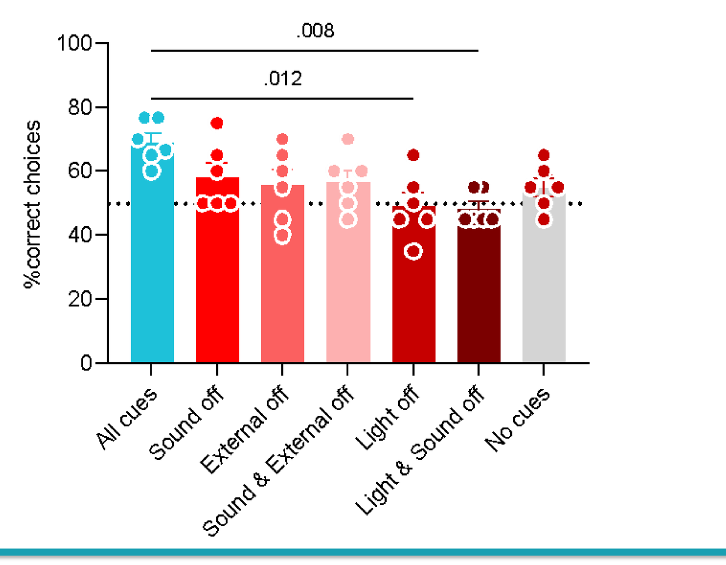
RESULTS

1. PSD-95-dependent activity of CA1 synapses, regulates decision-making strategies and attention.

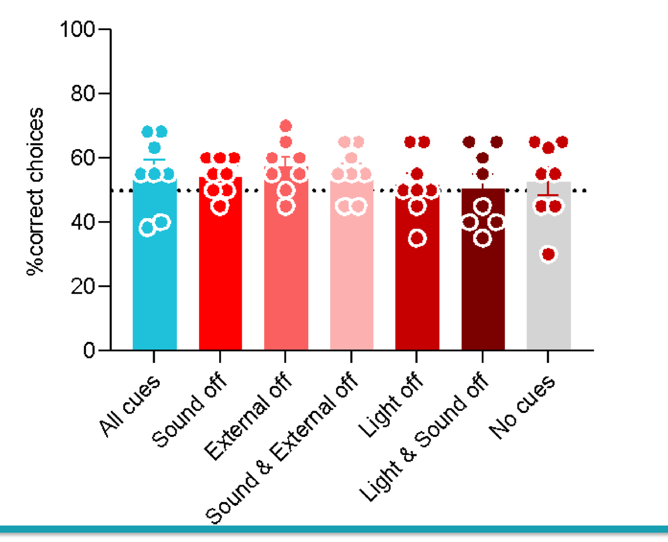


2. PSD-95 knockdown mice show cue-independent performance, suggesting impaired cue processing.

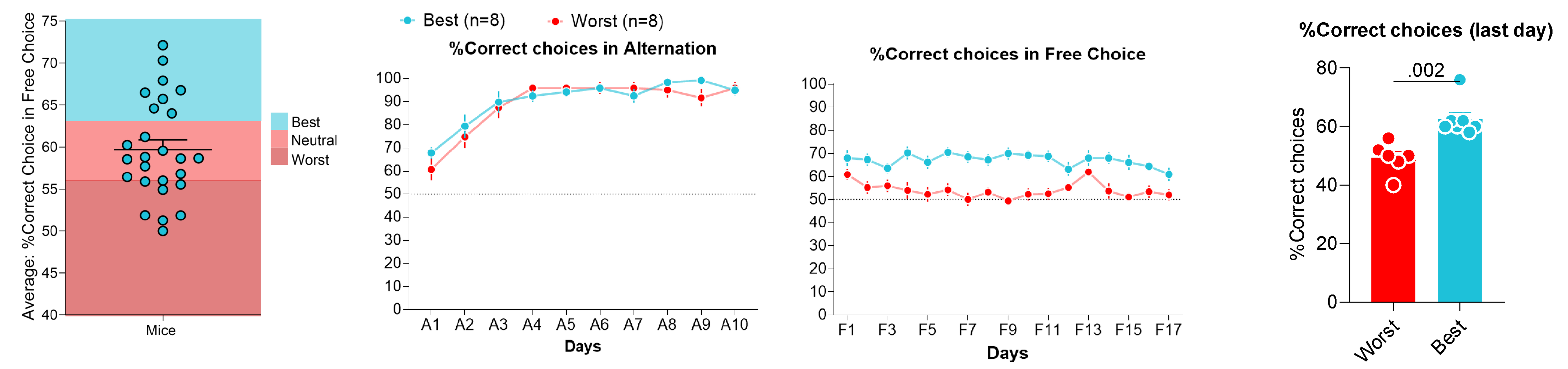
shRNA-luciferase injected

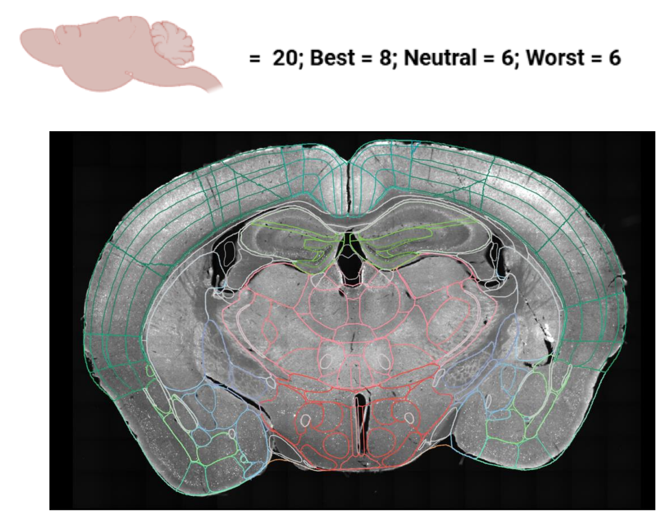


shRNA-PSD95-UB_GFP injected

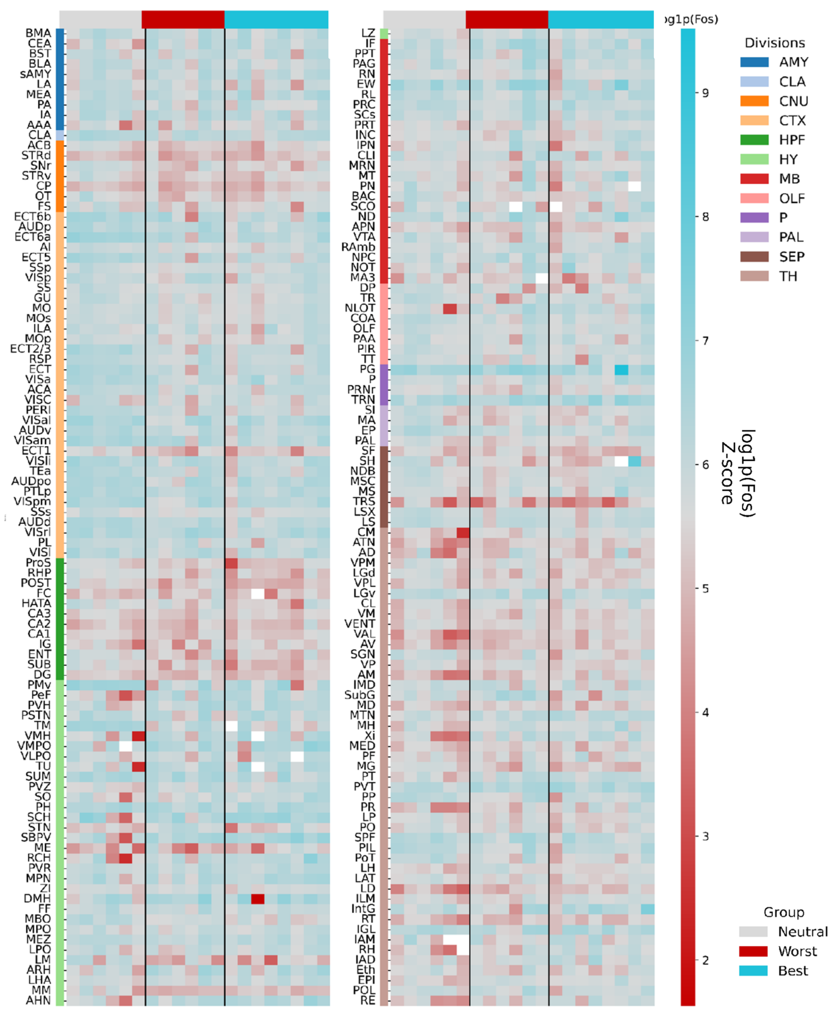


3. Mice exhibit individual differences in task performance. Brain-wide c-Fos activity reveals distinct network organization associated with accuracy of spatial choice.

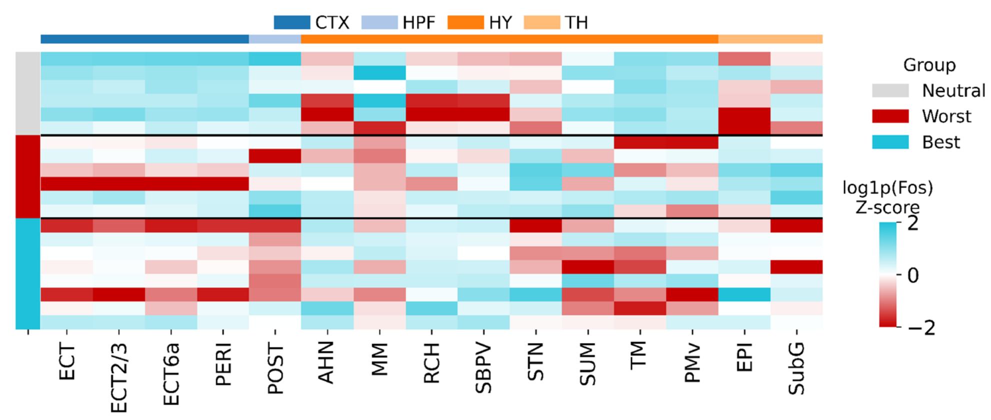




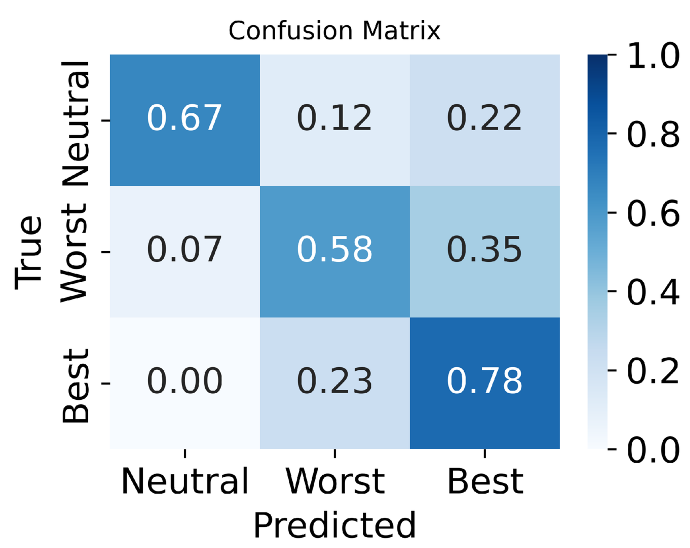
Whole brain activity



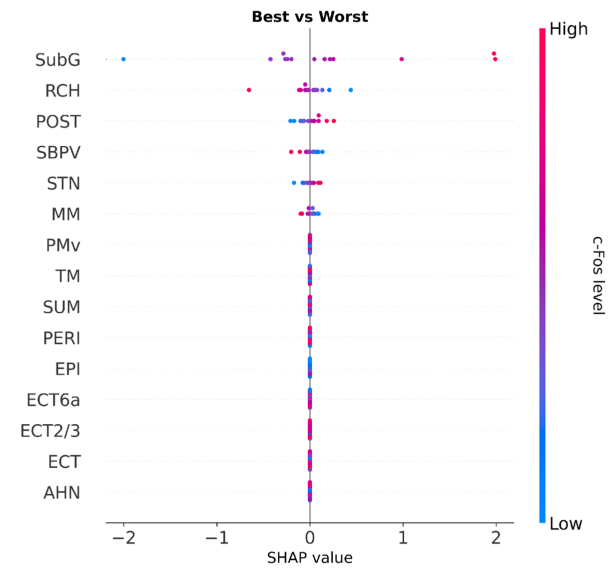
Boruta selected structures

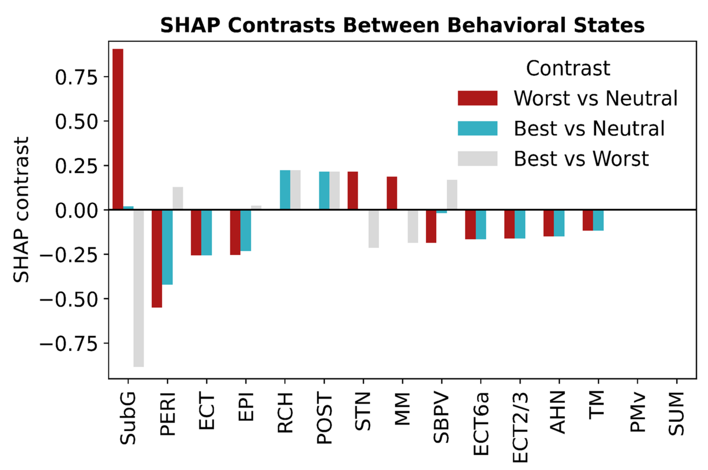


Model Perfomance

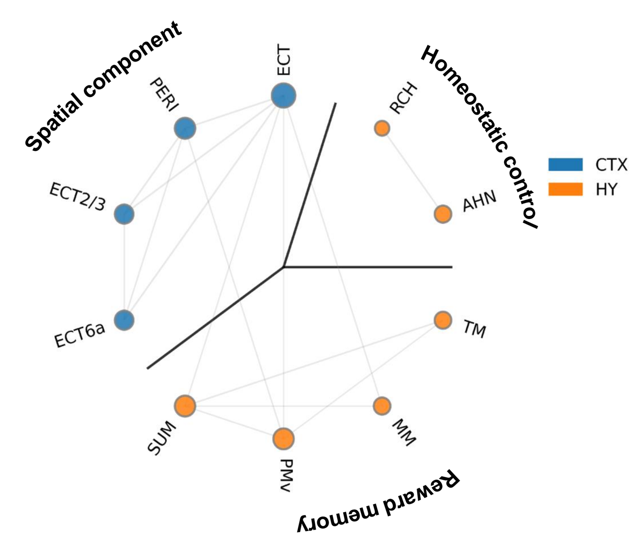


SHAP Values

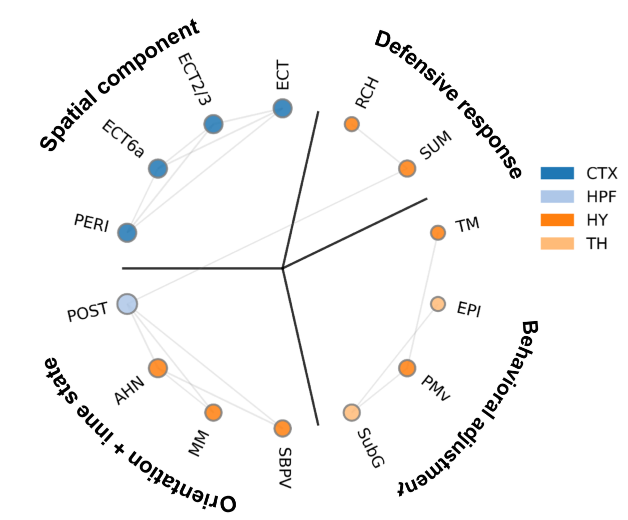




Best animals



Worst animals



CONCLUSIONS

1. PSD95-dependant synaptic plasticity in dCA1 regulates decision-making dynamics & attention.
2. Plasticity within dCA1 is neccessary for cue processing & navigation.
3. Successful animals exhibit more efficient spatial & reward networks organization, and unsuccessful animals show greater involvement of defense-related and regulatory circuitry.

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